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Thapar Institute of Engineering & Technology, Patiala, Punjab

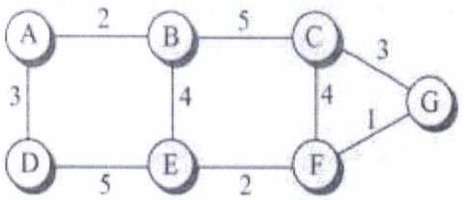
Department of Computer Science and Engineering

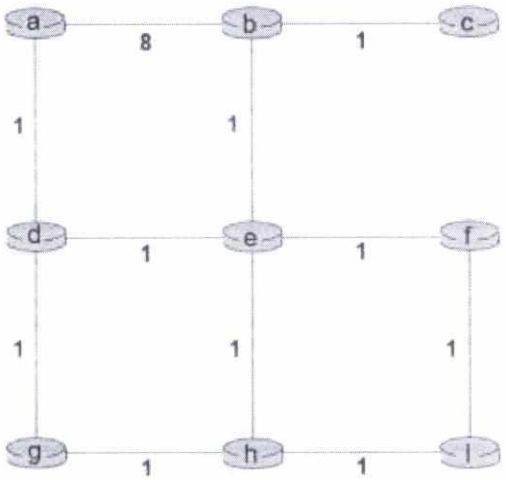
B.E. EST Examination (Summer Semester), July 2026

Course Code: UCS520 & UCS414	Course Name: Computer Networks
Date: 17th July, 2026	Time: 2:00-5:00 (Evening)
Duration: 3 Hours, Max. Marks: 40	Name of Faculty: Dr. Himanshu Gauttam

Note: (i) Attempt any three questions. One Mark will be awarded for attempting the questions (and subparts) in the order.

(ii) Assume missing data if any.

Q. No.	Questions	Marks	CLO
Q.1	(A)The graph represents the network of an organization, where node *A* generates the traffic and transmits to node *G*.  a. The weighted graph I. Determine the shortest path between node A and G using link state routing algorithm with key intermediate cost updates and total cost of final path. II. What will be the Link State Data Base entry at Node D.	(6)	L4
	(B) Explain the purpose and working of a DNS server.	(2)	L2
	(C) In a communication system, sender and receiver adopts the *CHECKSUM* scheme to detect the error. Describe the error detection procedure using the checksum method with the help of a suitable illustration.	(5)	L3
Q.2	(A) Explain the header format of an IPV4 packet.	(4)	L2
	(B) The Go-back-N ARQ protocol makes the following claim: "***The size of sender window is always less than 2^m (m is the size of the sequence number field in bits)***". Prove the validity of the same using a suitable example.	(5)	L4
	(C) Identify the limitation of traditional routing algorithms in the context of sustainable infrastructure. Discuss the key techniques to address the limitations.	(3)	L2
	(D) What is the key difference between Random Access and Reservation based MAC protocols.	(1)	L1

Q.3	(A) Discuss the concepts and logical steps involved in the data transmission for the circuit switched and packet switched networks with the help of suitable diagrams. Compare the performance of these techniques in terms of delay and efficiency.	**(6)**	**L3**
	(B) Explain the closed loop-based congestion control schemes.	**(4)**	**L3**
	(C) Identify and list out at least three key differences between UDP and TCP.	**(1.5)**	**L2**
	(D) A non-periodic composite signal has a bandwidth of 300 kHz, with a middle frequency of 180 kHz and peak amplitude of 10 V. The two extreme frequencies have an amplitude of 0. Draw the frequency domain of the signal.	**(1.5)**	**L2**
Q.4	(A) Explain the procedure of updating the distance vector(s) in Distance Vector Routing (DVR) algorithm with the help of a suitable diagram.	**(3)**	**L3**
	(B)  For the given network: I. At time t_1 node 'b' receives the distance vectors from node 'a', 'c' and 'e', respectively. Determine the updated distance vector at node 'a' by describing the key intermediate steps using DVR. II. Determine the distance vector of node 'c' at time t_2 (where $t_2 > t_1$) using DVR.	**(5)**	**L4**
	(C) Station X comprises a message having 9 frames, which needs to be sent to station Y using GO-BACK-N-ARQ protocol with window size 3. All frames are readily available for transmission. If every 5th frame transmitted by station X is lost but no ACK from station Y is lost, determine the total number of frame transmissions required for Station X to successfully deliver the entire message to Station Y.	**(5)**	**L3**